

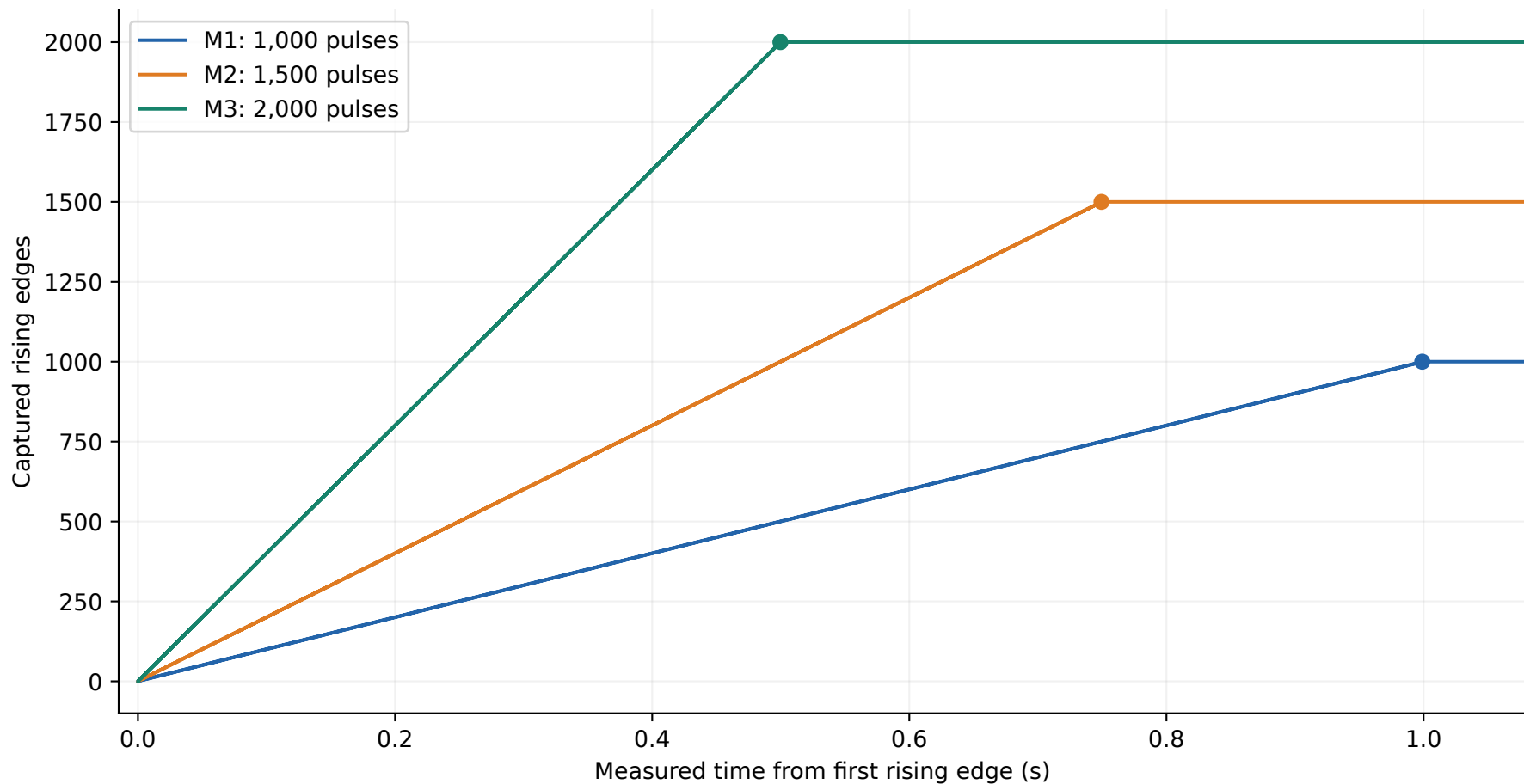
STEP hardware validation — pulse counts and stopping

Current firmware: reported pulse counts match the independent logic-analyzer captures.

Nominal rates within $\pm 0.1\%$; long-run residual drift is documented in figure 8.

Case	Captured M1/M2/M3	Reported M1/M2/M3	State	Counts
1	1/2/3	1/2/3	Completed	MATCH
2	1000/1500/2000	1000/1500/2000	Completed	MATCH
3	100000/100000/100000	100000/100000/100000	Completed	MATCH
4	1024/1024/1024	1024/1024/1024	Completed	MATCH
5	995/498/249	995/498/249	Stopped	MATCH
6	1000/1000/1000	1000/1000/1000	Completed	MATCH
7	512/256/128	512/256/128	Underrun	MATCH
8	435/435/435	435/435/435	Completed	MATCH

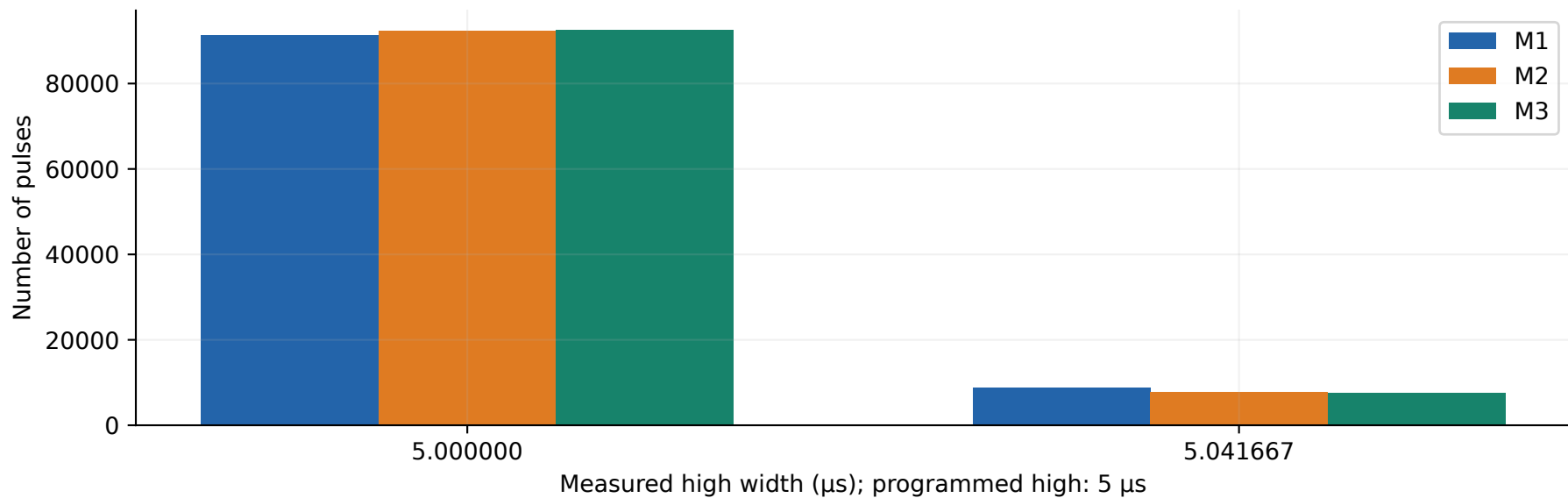
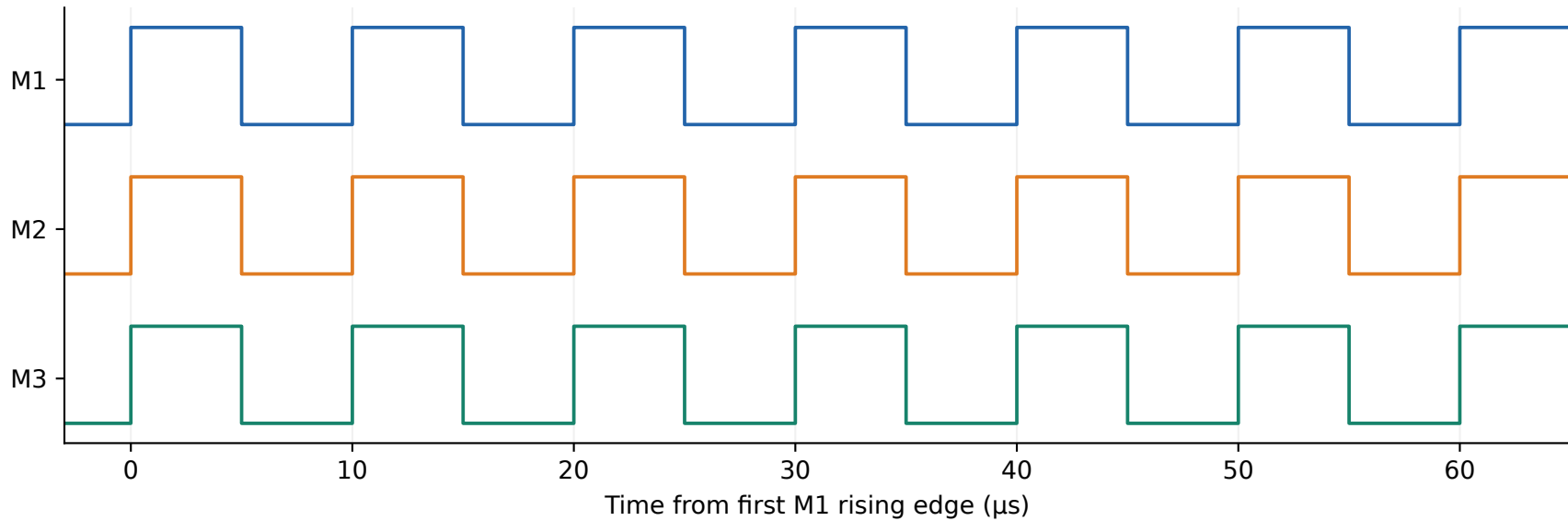
Case 2 — independent rates and finish times



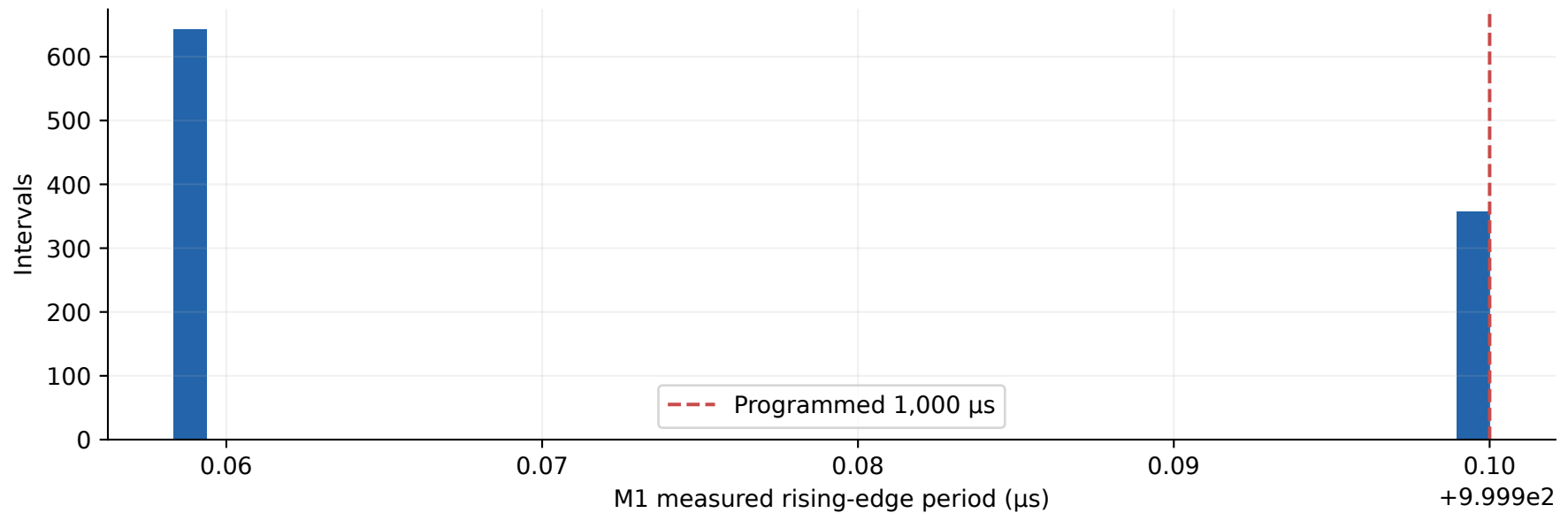
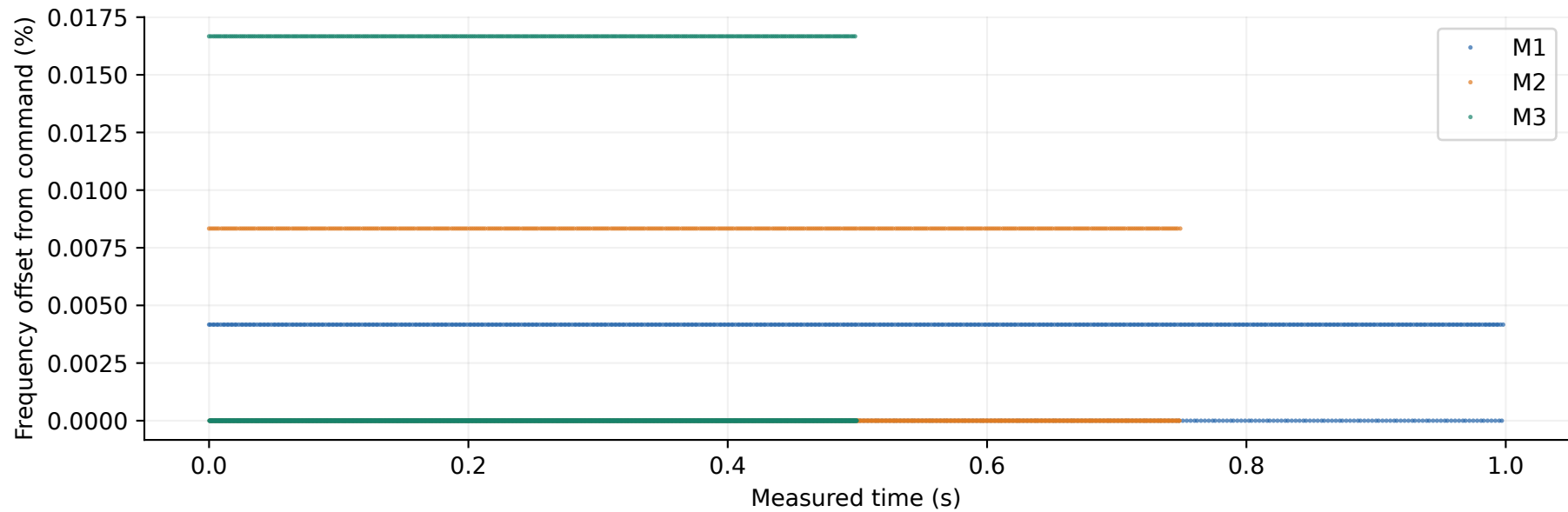
All first edges align within one 41.7 ns sample. Dots mark the final falling edges; no subsequent pulses in the capture.

Case 3 — 100,000 pulses/axis; measured M1 rate 100.003 kHz

First pulses: all three axes share the first edge

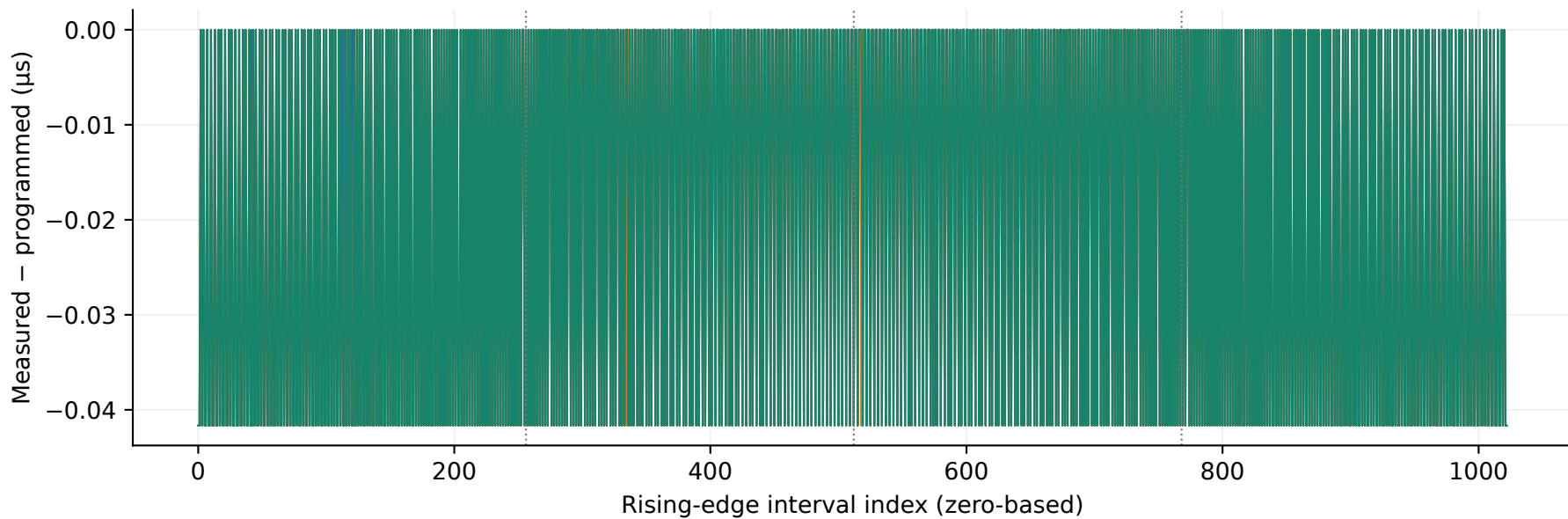
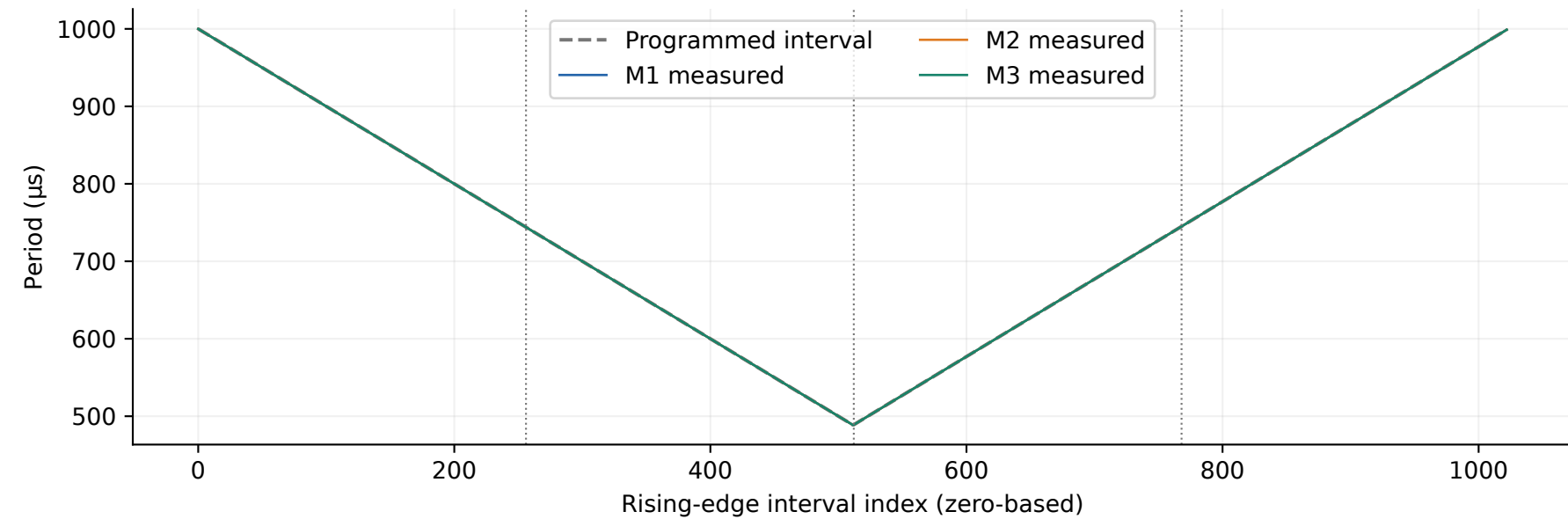


Measured frequency offset and interval distribution



Timing is relative to the analyzer clock. Sample quantization remains visible; instrument timebases are not calibrated.

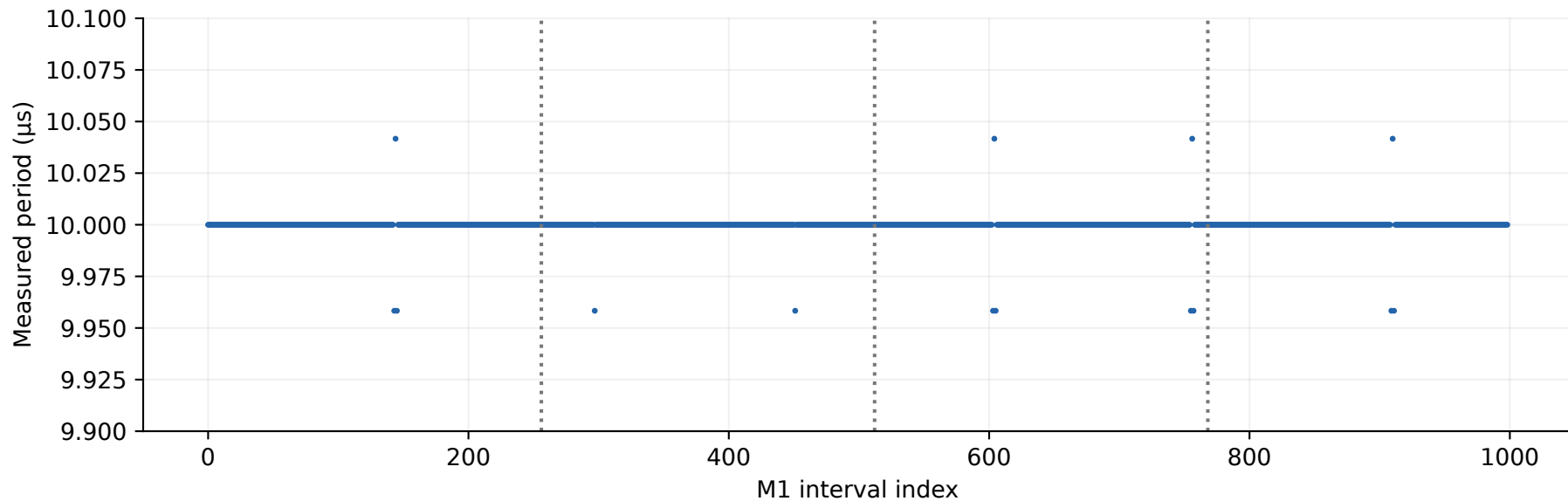
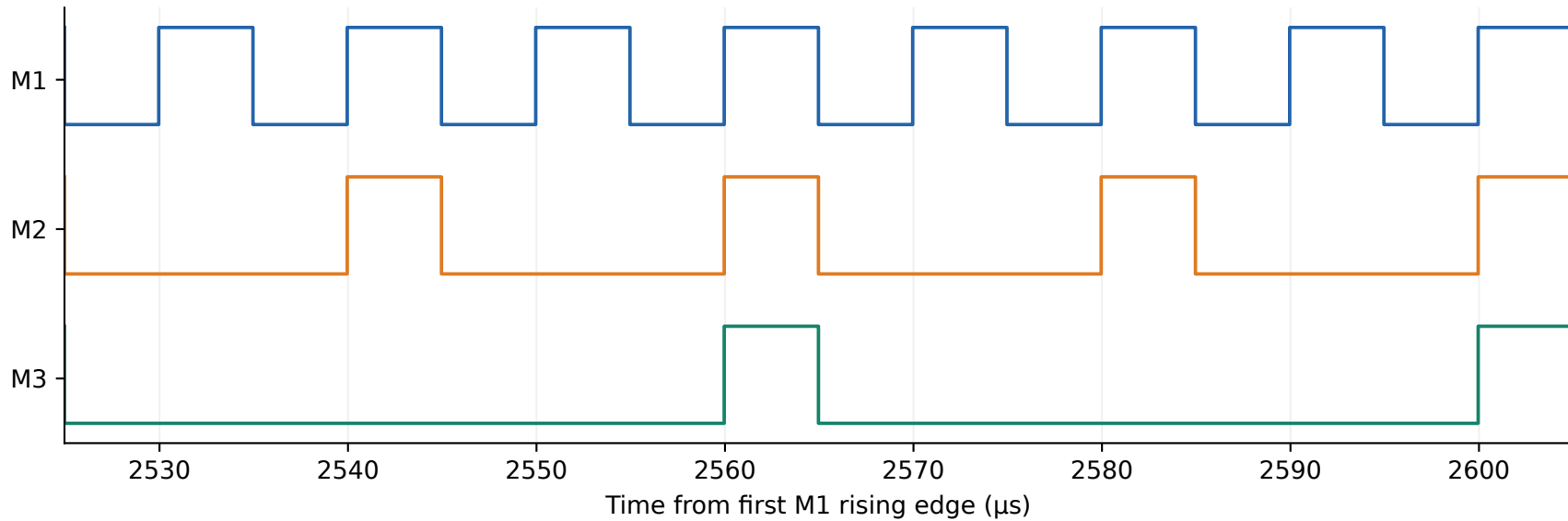
Case 4 — copied acceleration/deceleration profile, 1,024 pulses/axis



Dotted lines indicate 256-pulse buffer boundaries. Measured intervals are shown without clock correction.

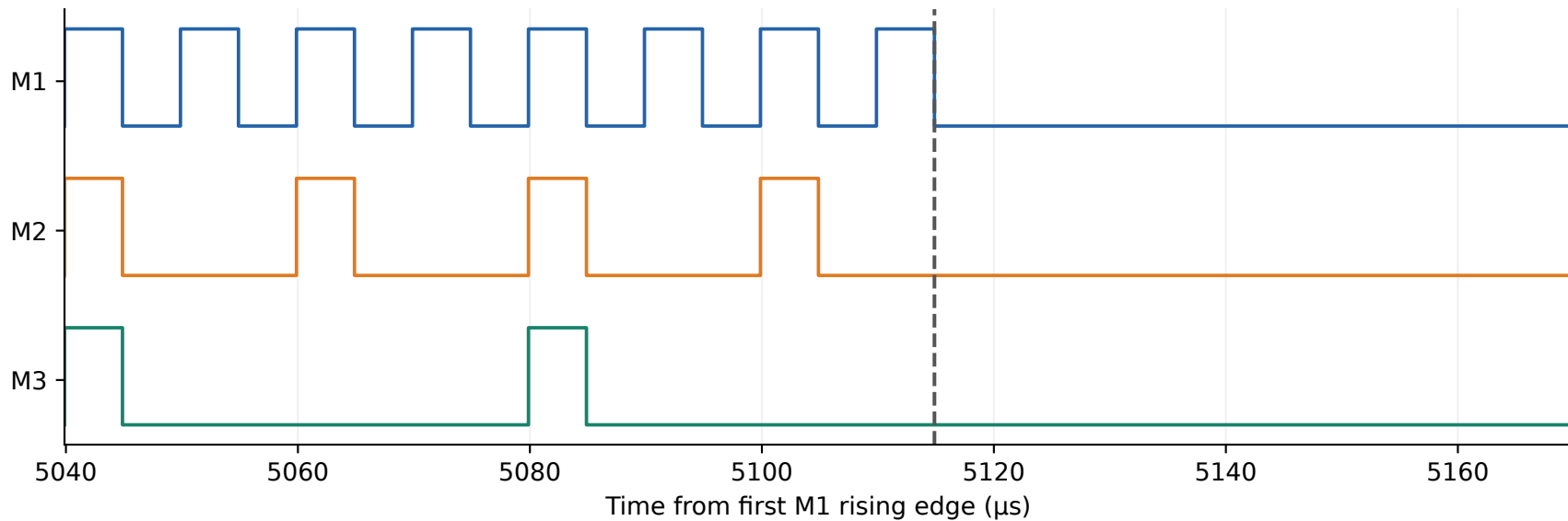
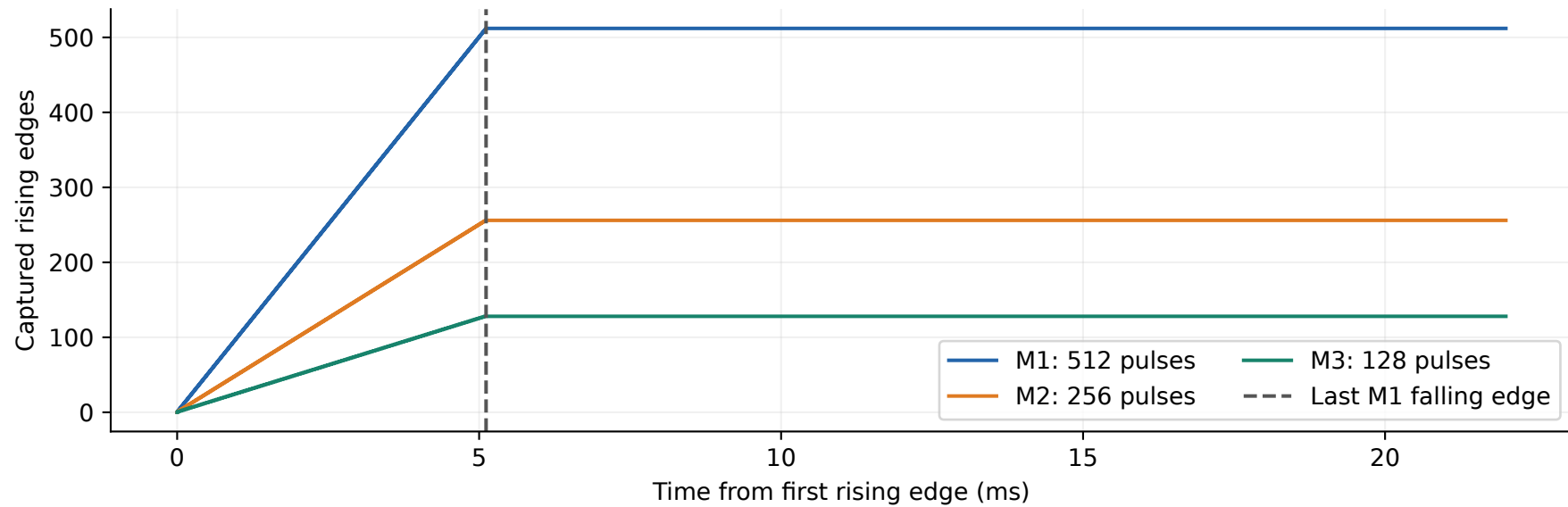
Case 6 — delayed refill IRQ: 1,000 captured pulses on every axis

Pulse train crosses the first M1 DMA-buffer boundary while its IRQ is withheld



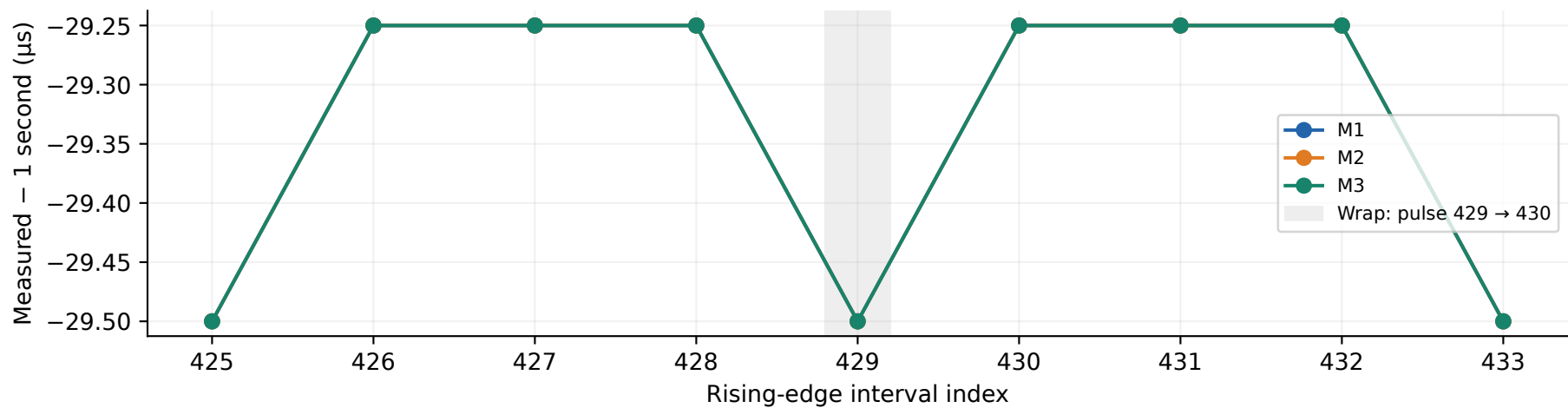
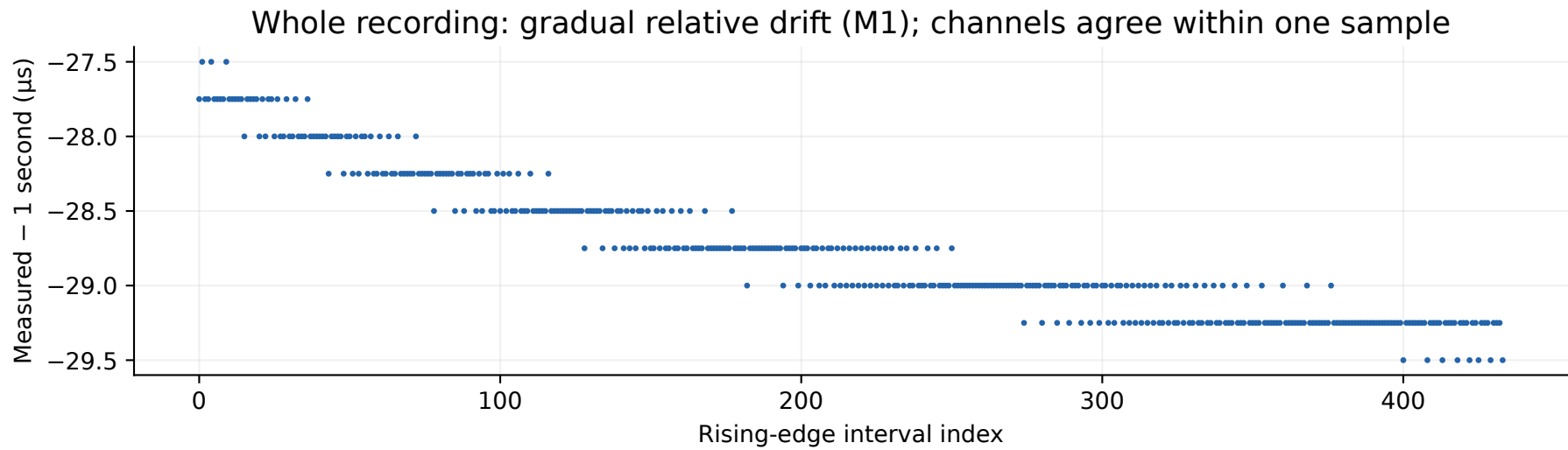
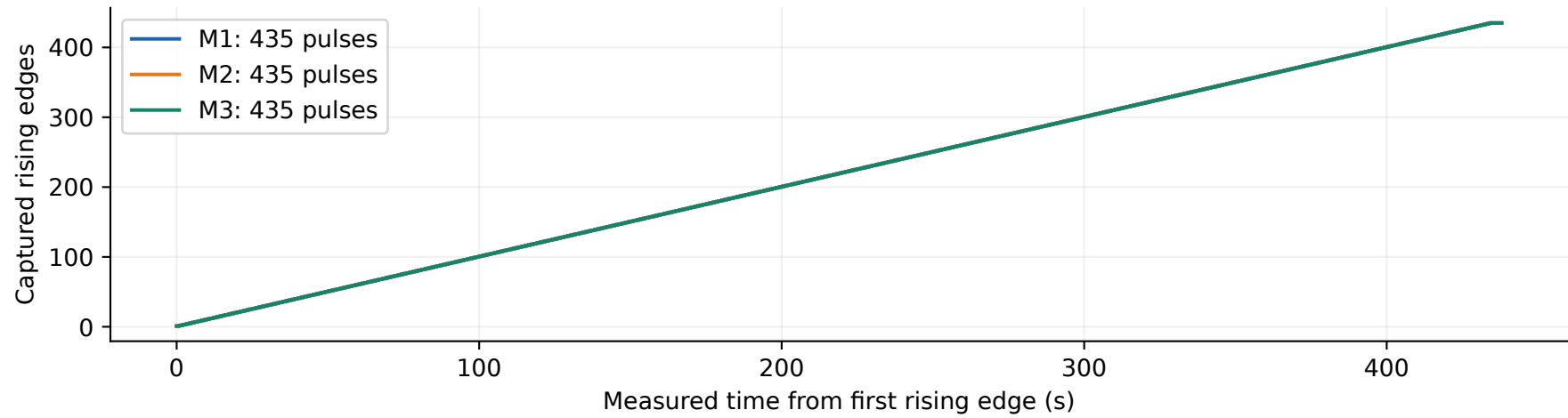
The firmware masks DMA IRQs for about 4.5 ms including the start delay. The analyzer observes the outputs, not the IRQ mask directly.

Case 7 — autonomous stop with all DMA IRQs withheld

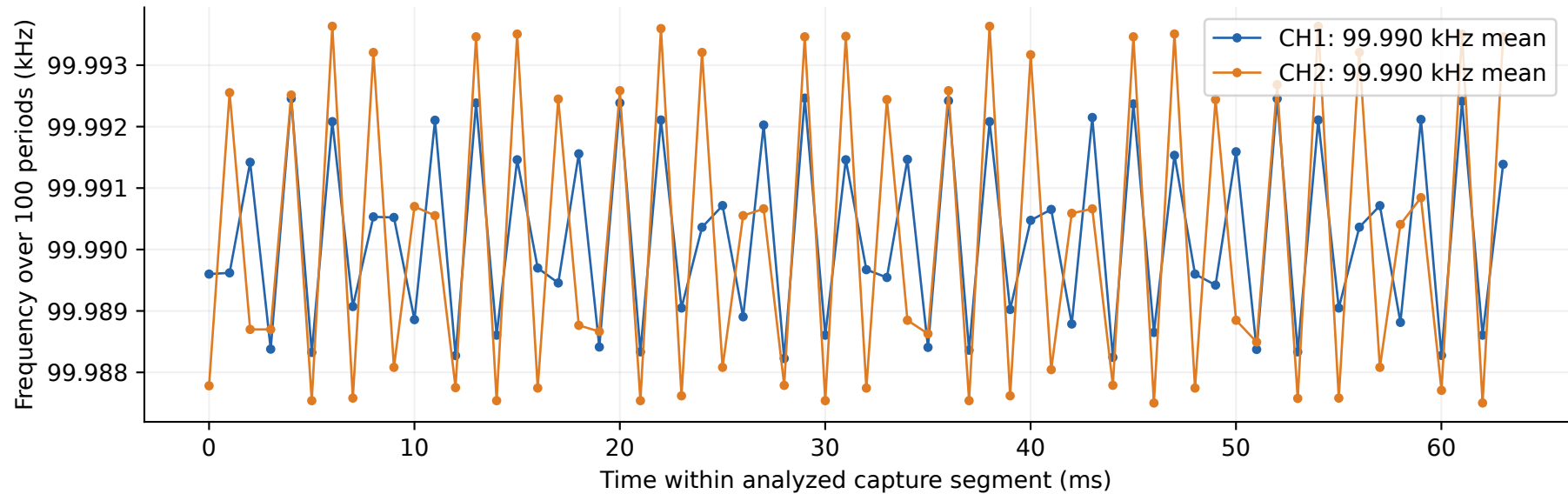
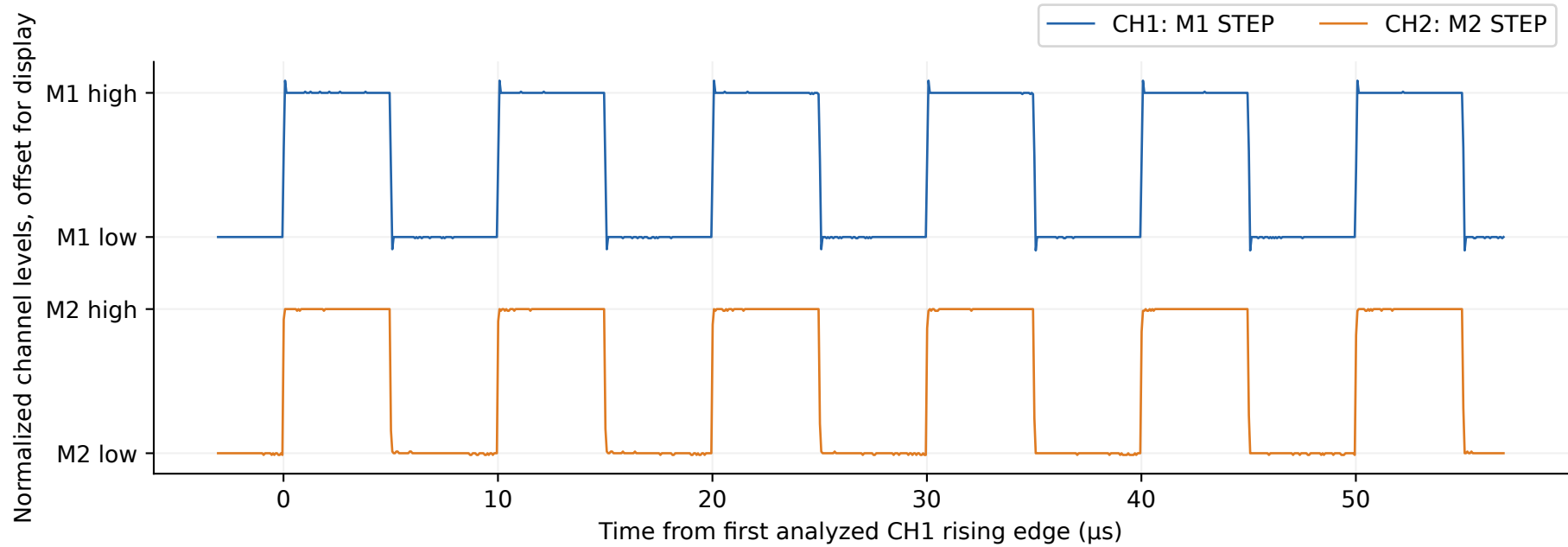


UART: Underrun, exact=1, counts=512/256/128, guard_stopped=1 before IRQ restoration. Outputs remain low after the final pulses.

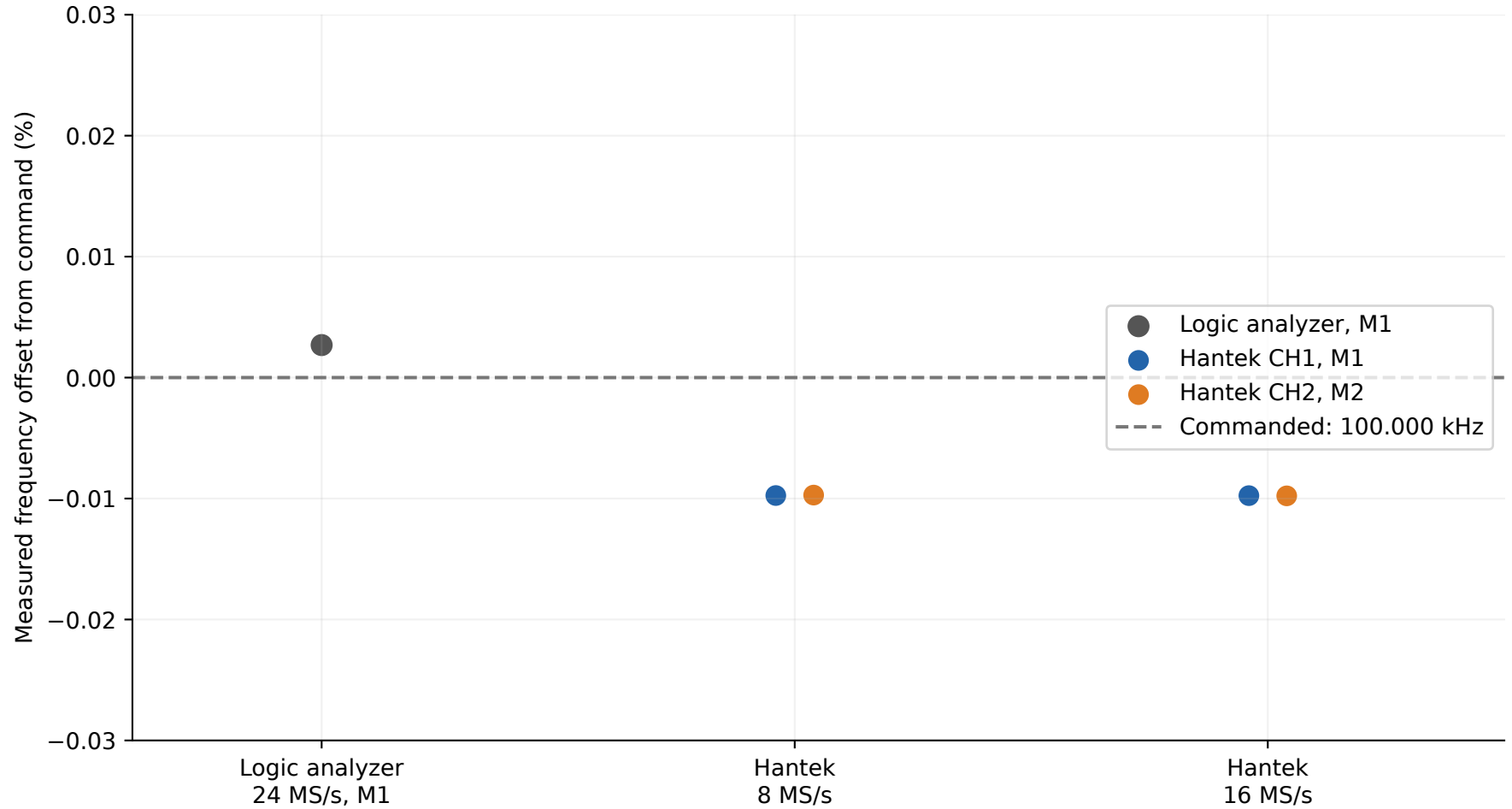
Counter rollover — complete pulse counts, with slow relative clock drift



Hantek 6022BE — simultaneous M1 and M2 STEP measurements

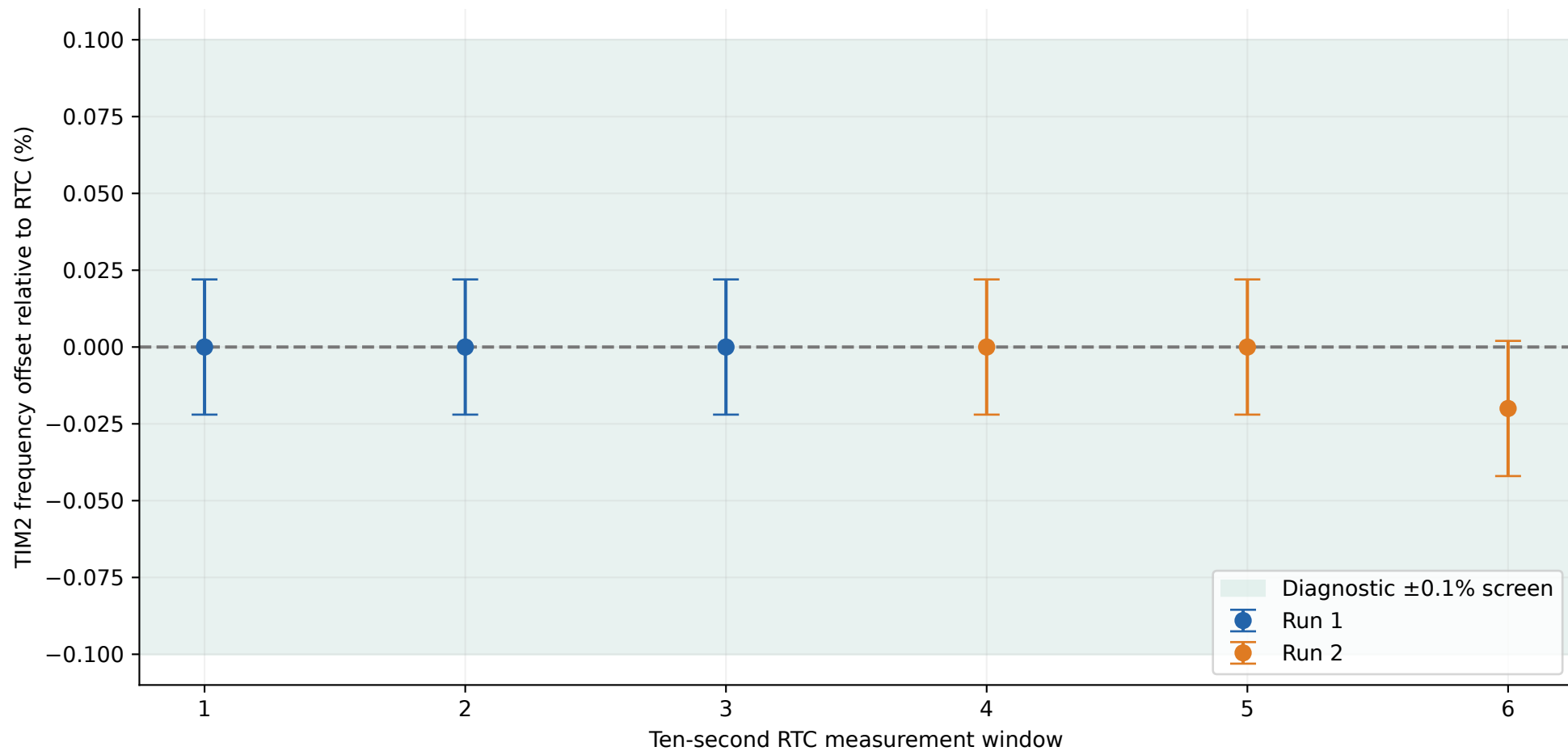


Independent STEP frequency measurements



Same firmware, separate bursts. Channel pairs are simultaneous within each Hantek capture. No instrument has an independent timebase calibration.

TIM2 measured against the independent RTC crystal



No external instrument. STEP held low. Error bars cover polling and synchronization; LSE crystal tolerance is additional.